In [1]:
```
import pandas as pd

df = pd.read_csv('../data/titanic.csv')

df.head()
```

Out[1]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [2]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [3]:
```
df.describe()
```

Out[3]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.91
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [4]:
```
df.isnull().sum()
```

Out[4]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [5]:
```
# Fill missing Age with median
df['Age'].fillna(df['Age'].median(), inplace=True)

# Fill missing Embarked with most frequent value
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Drop Cabin column (too many missing values)
df.drop(columns=['Cabin'], inplace=True)

# Check again
df.isnull().sum()
```

```
C:\Users\Madhu\AppData\Local\Temp\ipykernel_26384\2141042876.py:2: FutureWarning:
A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work becau
se the intermediate object on which we are setting values always behaves as a cop
y.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.meth
od({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to pe
rform the operation inplace on the original object.


  df['Age'].fillna(df['Age'].median(), inplace=True)
C:\Users\Madhu\AppData\Local\Temp\ipykernel_26384\2141042876.py:5: FutureWarning:
A value is trying to be set on a copy of a DataFrame or Series through chained as
signment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work becau
se the intermediate object on which we are setting values always behaves as a cop
y.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.meth
od({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to pe
rform the operation inplace on the original object.


  df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

Out[5]:

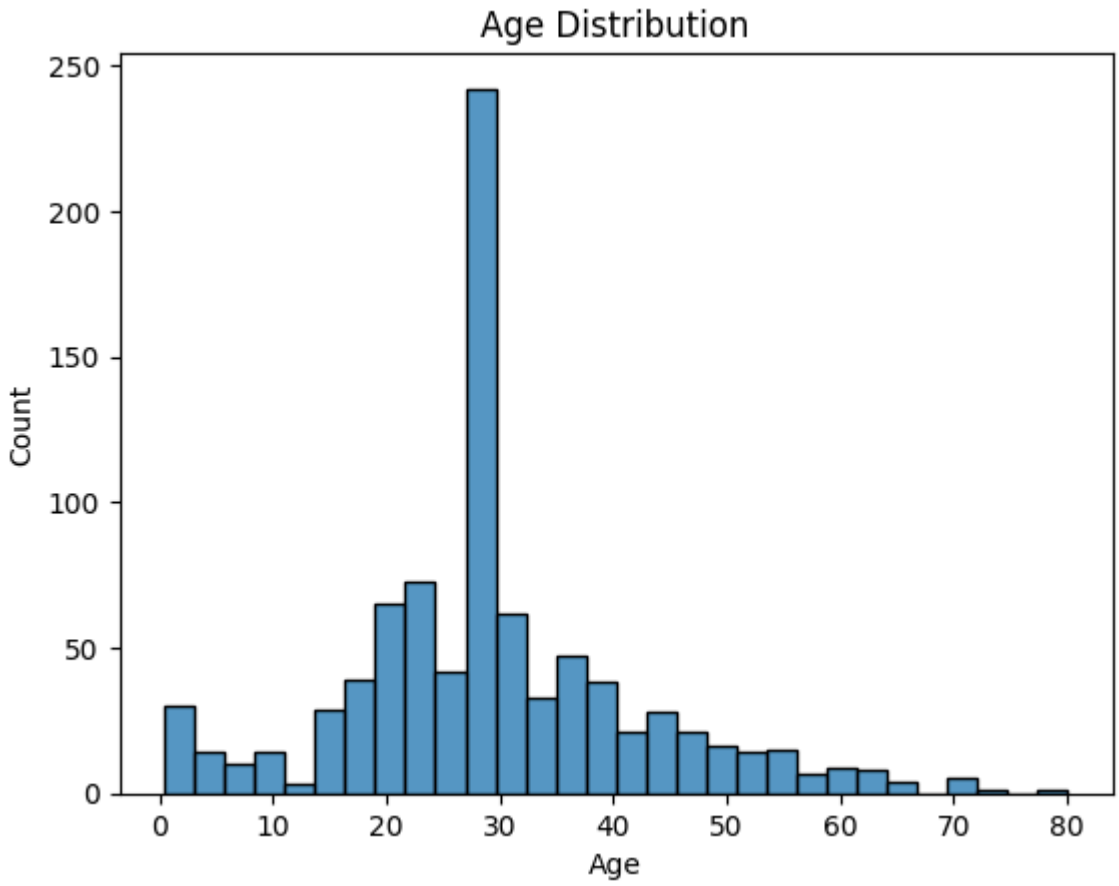
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

In [6]:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Age Distribution
plt.figure()
sns.histplot(df['Age'], bins=30)
plt.title("Age Distribution")

plt.savefig('../images/age_distribution.png')
plt.show()
```

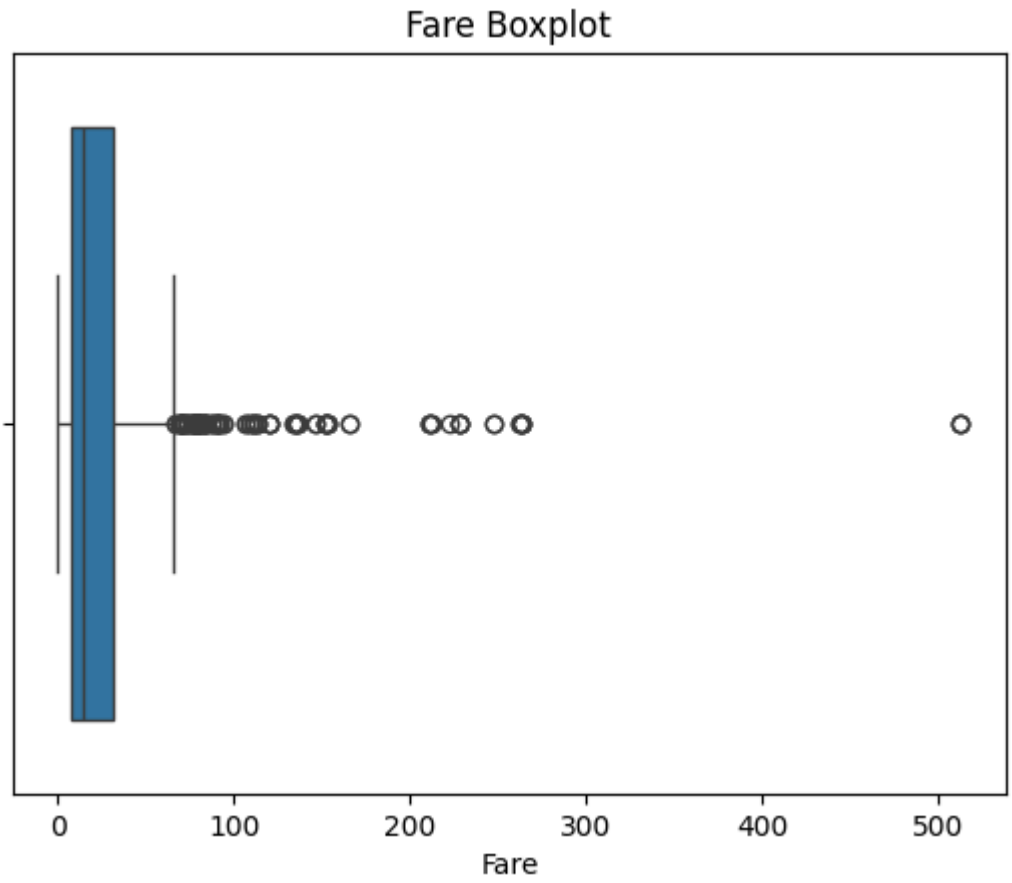


Observation:

- Most passengers are between 20–40 years
- Very few elderly passengers

In [7]:
```
# Fare Boxplot
plt.figure()
sns.boxplot(x=df['Fare'])
plt.title("Fare Boxplot")

plt.savefig('../images/fare_boxplot.png')
plt.show()
```

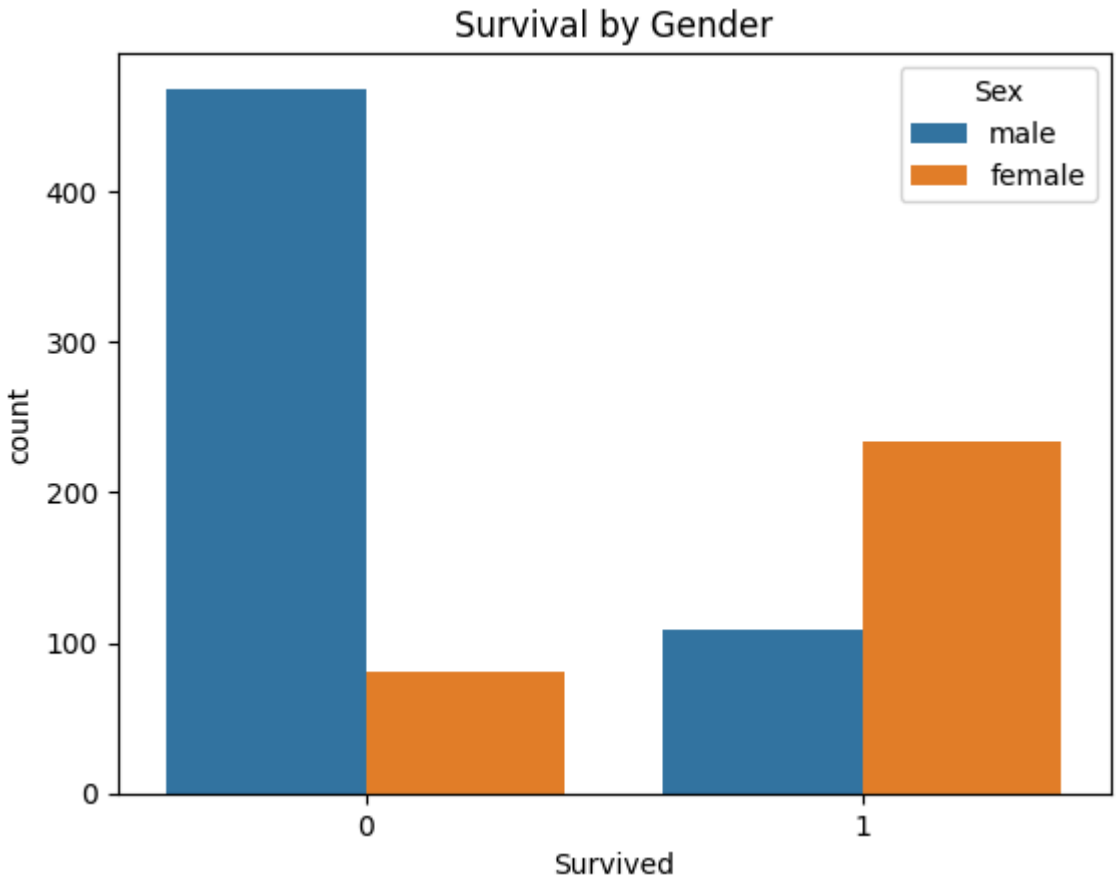


Observation:

- Fare has many outliers
- Some passengers paid very high fares

In [8]:
```
# Survival by Gender
plt.figure()
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival by Gender")

plt.savefig('../images/survival_gender.png')
plt.show()
```

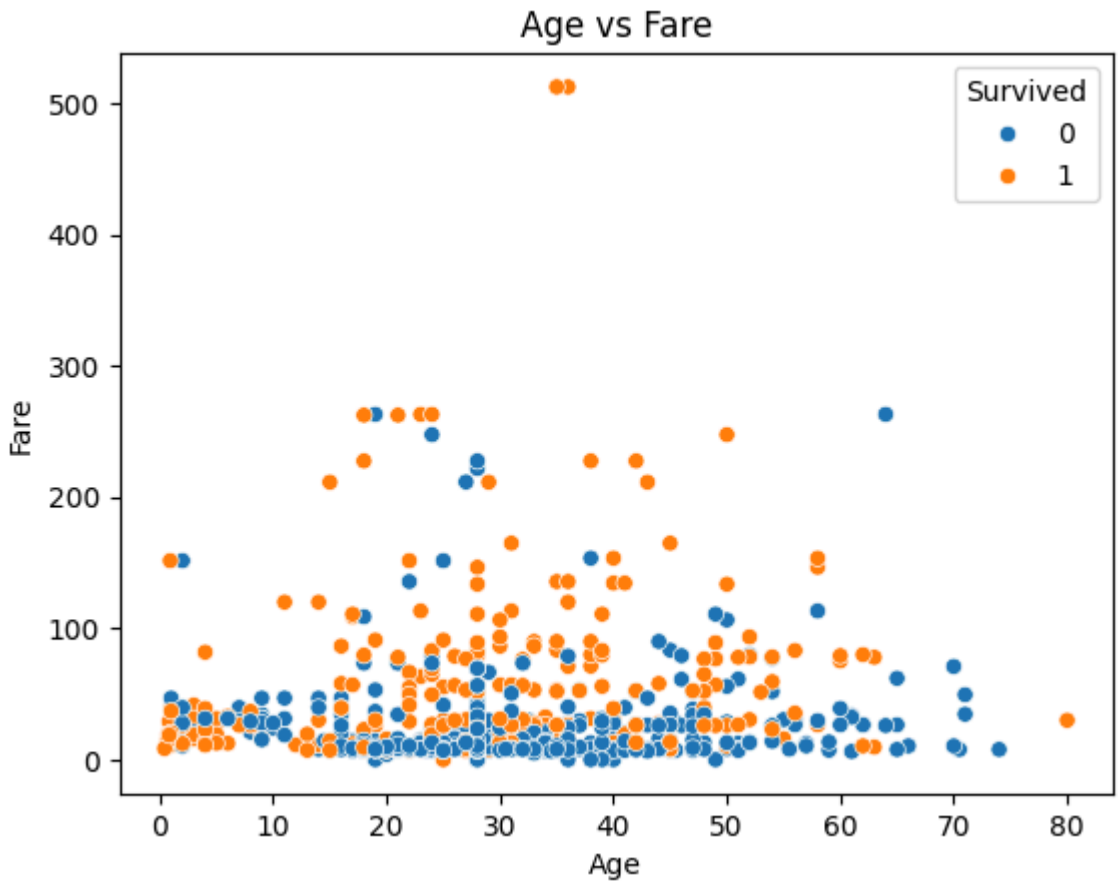


Observation:

- Females survived more than males
- Males had lower survival rate

In [9]:
```
# Age vs Fare
plt.figure()
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare")

plt.savefig('../images/age_fare.png')
plt.show()
```



Observation:

- Higher fare passengers had higher survival chances
- No strong pattern between age and fare

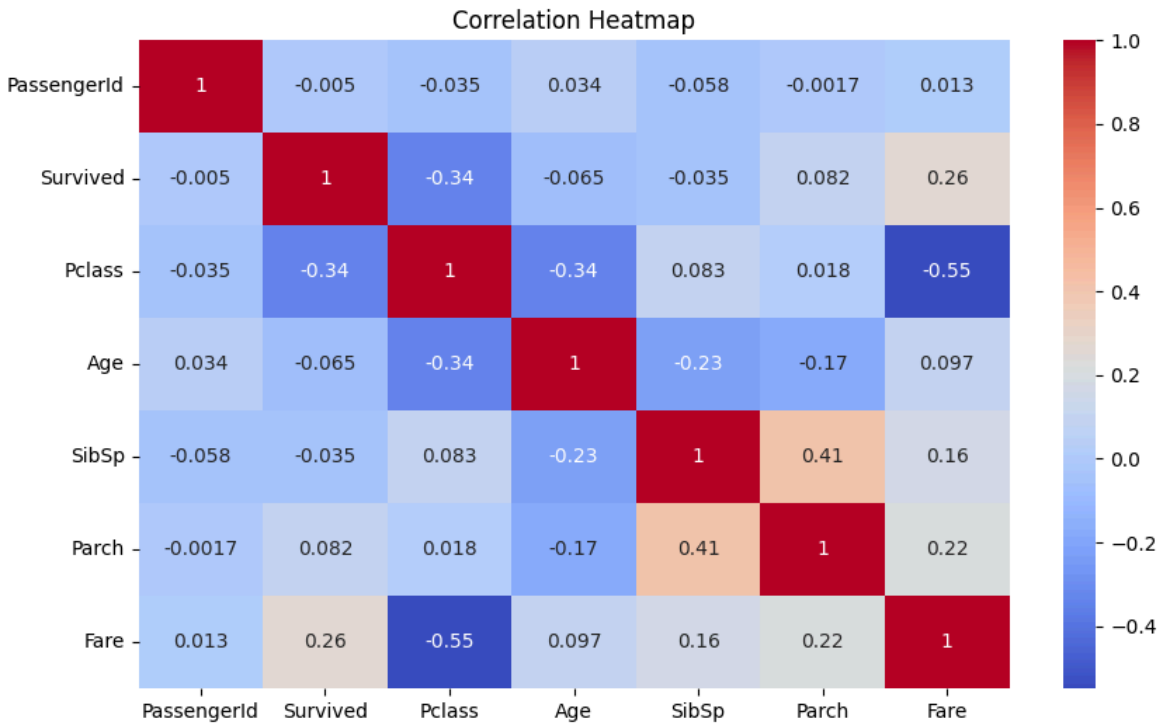
In [11]:
```
# Correlation Heatmap (FIXED)
plt.figure(figsize=(10,6))

numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.savefig('../images/heatmap.png')
plt.show()
```



Observation:

- Pclass negatively correlated with survival
- Fare positively correlated with survival
- Age has weak correlation with survival

Final Summary:

- Female passengers had higher survival rate than males
- Passengers in higher class (Pclass 1) had better survival chances
- Fare is positively related to survival (higher fare → more survival)
- Age has only slight impact on survival
- Most passengers were between 20–40 years
- Fare contains many outliers

Conclusion: Survival depended mainly on gender, class, and fare rather than age.